

UI/UX DESIGN SPECIFICATION

Apollo & LayerSignal Unified Dashboard

A comprehensive design specification for the next-generation trading dashboard that unifies Apollo backtesting and LayerSignal live scanning into a single, information-dense interface.

101

FEATURES

7

TABS

26

SQLITE TABLES

2

ENGINES

Design Philosophy

The Apollo-LayerSignal Unified Dashboard is designed for a power user who needs to process hundreds of data points across thousands of stocks in real time. Every design decision prioritizes rapid signal recognition over aesthetic minimalism. The interface must feel like a command center, not a marketing page. The following five principles guide every layout, color, and interaction choice in this specification.

PRINCIPLE 01

Progressive Disclosure, Not Progressive Reduction

The dashboard shows maximum relevant information at the top level. Detail is revealed through expand, hover, and drill-down rather than being hidden behind navigation. A trader scanning 340 stocks in the Screener tab should see every signal-bearing column without clicking. Rows expand in-place for pool breakdowns and gate details. The philosophy is: show everything that matters, collapse only what is context-dependent.

PRINCIPLE 02

Signal-First Visual Grammar

Every data point is encoded as a visual signal: badges for states (Entry, Exit, Hold), color-coded RSI values, micro-dot gate indicators, and proximity bars for 52-week positioning. The eye should parse the table row in under 2 seconds without reading a single number. This means badges, color fills, and positional encodings take precedence over raw numeric displays. Numbers appear on hover or in expanded rows.

PRINCIPLE 03

Zero-Context Comprehension

A new user opening any tab should understand what they are looking at within 10 seconds. Every column header has a tooltip explaining its origin (Apollo, LayerSignal, or computed). Every badge is self-labeling. The color system maps to a fixed semantic vocabulary: green for bullish signals, red for bearish, amber for caution, blue for informational, purple for Apollo-specific, teal for system status. No legend should be required to interpret the primary view.

PRINCIPLE 04

Spatial Proximity of Related Signals

Data that is analyzed together must be displayed together. In the Watchlist, the three-column layout places the stock list, primary analysis, and deep-dive accordion panels side by side. The RSI Stack, RSI Cushion, and RSI Comparison accordion sections are adjacent because a trader evaluates them as a group. This principle rejects the common pattern of burying related data across multiple pages or modal dialogs.

PRINCIPLE 05

Engine Neutrality with Unified Context

The user should not need to know whether a signal came from Apollo or LayerSignal. Both engines write to the same SQLite database. The dashboard reads from that database and presents a unified view. However, engine provenance is available on hover (via tooltip) and in the System tab for debugging. The tab structure is task-based (Screener, Watchlist, Scanner) rather than engine-based (Apollo, LayerSignal), a deliberate departure from the architecture document.

Information Architecture

The dashboard is organized into seven task-based tabs. Each tab maps to a distinct trader workflow. The priority levels (P0, P1, P2) determine the implementation order, with P0 tabs forming the minimum viable dashboard. A critical architectural decision was made: the tabs follow task-based organization, not engine-based workspaces. The user thinks in terms of "scan my universe" and "check my watchlist," not "use Apollo" and "use LayerSignal." This is a deliberate departure from the architecture document which proposed an engine-mode toggle.

Tab Structure

TAB	PRIORITY	PURPOSE	PRIMARY ENGINE
Screener	P0	Universe-wide signal scanning across all stocks	Both
Watchlist	P0	Monitored stock deep dives with full analysis	Both
Scanner	P1	Full market pattern scan with presets	LayerSignal
Analytics	P0	Backtest results, profiles, walk-forward validation	Apollo
Guidance	P1	Post-trade behavior analysis and recommendations	Apollo
Alerts	P0	Chronological cross-engine alert feed	Both
System	P2	Engine health, data quality, configuration	System

Navigation Hierarchy

App Bar -- Logo, Global Search, Alert Bell, Engine Health, Settings

Tab Bar -- 7 tabs (persistent navigation)

- Screener** -- Filter Sidebar | Universe Funnel | Stock Table | Expanded Rows
- Watchlist** -- Stock List | Center Panel (Chart/Analysis/Holdings) | Detail Accordion
- Scanner** -- Pattern Presets | Filter Sidebar | Result Table
- Analytics** -- Trade Engine | Profiles | Walk-Forward | Cross-Report | Fundamentals
- Guidance** -- Stock Selector | Behavioral Profile | Recommendations | MAE/MFE
- Alerts** -- Filter Bar | Chronological Alert Feed
- System** -- Engine Cards | API Endpoints | Score Distribution

Market Regime Banner -- BULLISH/NEUTRAL/BEARISH/VOLATILE + NIFTY + VIX

Content Area -- Tab-specific content renders here

Ticker Footer -- 14 scrolling index values

The task-based tab structure means a trader never has to think about which engine generated a signal. The Screener table seamlessly mixes Apollo's bucket classification and conviction scores with LayerSignal's RSI stacks and Renko signals. The Alerts feed interweaves entries from both engines chronologically. Only the System tab exposes engine internals, and only for debugging purposes.

Design System

The design system defines the visual vocabulary used across all 7 tabs. Every color, typeface, and component is chosen to maximize information density while maintaining legibility. The dark blue theme (not gray) was chosen to reduce eye strain during extended trading sessions and to provide a distinctive identity that separates this dashboard from generic gray-dark UIs.

Background Tiers (5 Levels)

Tier 0 (#0B1120) is the body background. Tier 1 (#111827) is the default card and panel background. Tier 2 (#1E293B) is for table headers, borders, and subtle elevation. Tier 3 (#334155) is for interactive elements, hover states, and wireframe borders. Tier 4 (#475569) is the lightest surface, used for disabled states and background fills.

Text Hierarchy (4 Levels)

LEVEL	COLOR	USAGE	CONTRAST RATIO
Primary	#F1F5F9	Headings, stat numbers, titles	15.4:1
Secondary	#CBD5E1	Body text, table cell content	10.2:1
Tertiary	#94A3B8	Labels, descriptions, metadata	6.1:1
Muted	#64748B	Timestamps, micro-labels, hints	3.8:1

Semantic Signal Colors

Green Bullish

Red Bearish

Amber Caution

Blue Info

Purple Apollo

Teal System

The RSI indicator colors are locked from LayerSignal v21 and must not be changed: RSI21 displays in green, RSI36 in blue, and RSI56 in amber. These colors are burned into trader muscle memory from months of LayerSignal usage and any change would cause confusion. The purple color is reserved for Apollo-specific elements (backtest profiles, walk-forward results, conviction scores) to provide subtle engine provenance without breaking the unified interface.

Typography Scale (10 Roles)

ROLE	FONT	SIZE	WEIGHT	USAGE
Stat Number	JetBrains Mono	22-28px	700	Dashboard stat cards, large metrics
Chapter Title	Inter	20-24px	800	Tab titles, chapter headings
Section Heading	Inter	14-16px	700	Sub-section titles within tabs
Sub-heading	Inter	12-13px	600	Card titles, group labels
Body Text	Inter	11px	400	Paragraph text, descriptions
Table Cell	Inter	10-10.5px	400	Table data cells
Badge / Chip	Inter	8.5-9px	600	Status badges, filter pills
Data Value	JetBrains Mono	11-13px	500	Numeric values in tables and cards
Table Header	Inter	8-8.5px	600	Column headers, uppercase
Micro Label	Inter	9px	500	Timestamps, tiny metadata

Component Library (11 Components)

COMPONENT	DESCRIPTION	USED IN
Badge	Color-coded pill showing state (Entry, Exit, Hold, L1-L4)	All tabs
FlagChip	Small chip displaying a binary flag with on/off styling	Screener, Watchlist
StatCard	Elevated card with large number, label, and optional delta	Watchlist, Analytics, System
MetricBadge	Compact badge pairing a label with a numeric value	Screener, Watchlist
ProximityBar	Horizontal bar showing position within a range (e.g., 52-week)	Screener, Watchlist
Sparkline	30-day miniature line chart embedded in table rows	Screener, Watchlist list
MiniGauge	Arc or radial gauge showing a percentage or score	System, Analytics
SortableTable	Full-featured table with column sorting, sticky header, expand rows	Screener, Scanner, Analytics
FilterPill	Toggle pill for quick filtering (active/inactive states)	Screener, Watchlist, Alerts

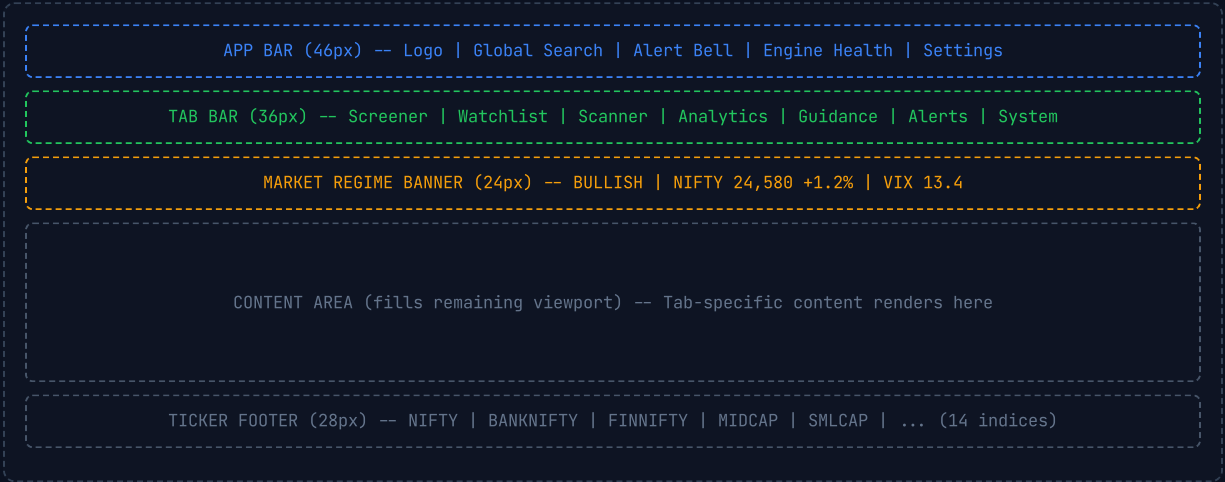
AccordionPanel	Collapsible section with chevron toggle and content area	Watchlist, Guidance
Tooltip	Hover-activated info panel showing definition and source engine	All tabs

CHAPTER 4

Global Shell Layout

The global shell wraps every tab in a consistent frame. From top to bottom, the shell consists of five zones: the App Bar, Tab Bar, Market Regime Banner, Content Area, and Ticker Footer. These zones are always visible regardless of which tab is active. The total chrome height is approximately 134 pixels (46 + 36 + 24 + 28 for the footer), leaving the content area to fill the remaining viewport height.

Shell Wireframe



App Bar Components

The App Bar is fixed at 46px height with a background of #0B1120 (deepest tier). It contains five elements from left to right: (1) the Apollo + LayerSignal logo mark, a compact icon that reinforces the unified brand; (2) a Global Search input that searches across symbols, features, and settings; (3) an Alert Bell icon with a red dot indicator showing unread alert count; (4) an Engine Health indicator showing green/amber/red dots for both engines; and (5) a Settings gear icon opening a preferences panel.

Market Regime Banner

The Market Regime Banner sits directly below the Tab Bar and spans the full width. It displays three pieces of information: the current market regime (BULLISH in green, NEUTRAL in blue, BEARISH in red, VOLATILE in amber), the NIFTY 50 index value with daily change percentage, and the India VIX value. The regime is computed by Apollo's regime detection module and stored in the cross_report table. This banner provides instant context for interpreting all signals below it.

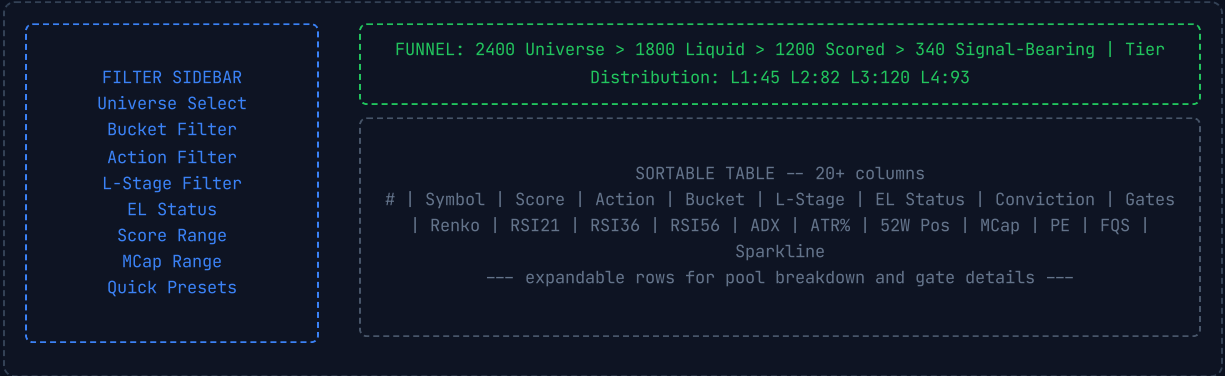
Ticker Footer

The footer is a 28px tall bar at the bottom of the viewport showing 14 scrolling index values: NIFTY 50, BANK NIFTY, NIFTY FIN SERVICE, NIFTY IT, NIFTY MIDCAP 100, NIFTY SMLCAP 250, INDIA VIX, NIFTY PHARMA, NIFTY AUTO, NIFTY METAL, NIFTY ENERGY, NIFTY PSU BANK, NIFTY REALTY, and NIFTY MEDIA. Values are pulled from the indices table in SQLite and updated by the NSE data feed engine. The ticker auto-scrolls horizontally when all 14 indices cannot fit in the viewport width.

Tab 1 -- Screener (P0)

The Screener is the primary landing tab and the highest-traffic view in the dashboard. It presents the entire tradable universe (approximately 2400 stocks) filtered through a multi-stage funnel down to the most actionable signals. The layout is a classic master-detail pattern: a 25% width filter sidebar on the left and a 75% width content area on the right containing the funnel summary and sortable stock table.

Screener Layout Wireframe



Universe Funnel Summary

Above the main table, a horizontal bar shows the four-stage universe funnel. Starting from the full universe (approximately 2400 stocks from the F&O segment and NSE broad market), the funnel shows how many stocks pass each filter stage: liquid stocks (those with sufficient trading volume, typically around 1800), scored stocks (those that pass Apollo's scoring pipeline, typically around 1200), and signal-bearing stocks (those with an active Entry, Exit, or Hold action, typically 200-400 depending on market conditions). Each stage is clickable to filter the table to that subset.

Adjacent to the funnel, a tier distribution summary shows the count of stocks in each LayerSignal bucket (L1 through L4). This gives an immediate sense of market breadth: a market with many L1 and L2 stocks suggests broad bullish momentum, while a market dominated by L4 stocks suggests defensive positioning is warranted.

Screener Table Columns (20+)

#	COLUMN	TYPE	SOURCE	DESCRIPTION
1	#	Index	UI	Row number for reference
2	Symbol	Text	Both	NSE trading symbol, bold, clickable to Watchlist
3	Score	StatCard	Apollo	Composite entry score (0-100), color-coded by tier
4	Action	Badge	Apollo	Entry (green), Exit (red), Hold (amber), None (muted)
5	Bucket	Badge	LayerSignal	L1 (green), L2 (blue), L3 (amber), L4 (red)
6	L-Stage	Text	LayerSignal	LayerSignal stage label from scan results
7	EL Status	FlagChip	Apollo	Entry Layer reached (EL1-EL4) or not
8	Conviction	MetricBadge	Apollo	Conviction score (0.0-1.0) with strength indicator
9	Gates	Micro-dots	Apollo	5 binary gate indicators as colored dots
10	Renko	Badge	LayerSignal	Current Renko brick color and trend
11	RSI21	Color value	LayerSignal	21-period RSI, colored green
12	RSI36	Color value	LayerSignal	36-period RSI, colored blue
13	RSI56	Color value	LayerSignal	56-period RSI, colored amber
14	ADX	Data	LayerSignal	Average Directional Index value

15	ATR%	Data	LayerSignal	ATR as percentage of price (volatility)
16	52W Pos	ProximityBar	Computed	52-week high/low position as horizontal bar
17	MCap	Data	Apollo	Market capitalization category (Large/Mid/Small)
18	PE	Data	Apollo	Price-to-earnings ratio from fundamentals
19	FQS	Badge	Apollo	Fundamental Quality Score (A/B/C/D)
20	Sparkline	Sparkline	LayerSignal	30-day price mini-chart

Expandable Rows

Clicking any row expands it in-place to reveal two sub-sections. The first is a Pool Breakdown showing which RSI pool the stock belongs to (based on RSI21/36/56 combinations) with a visual indicator of the pool's historical win rate. The second is Gate Details, showing the status of all 5 Apollo gates (regime, trend, momentum, volatility, quality) with pass/fail indicators and the specific values that determined each gate. This expansion avoids the need for a separate detail page and keeps the trader in the flow of scanning.

Feature-to-UI Mapping

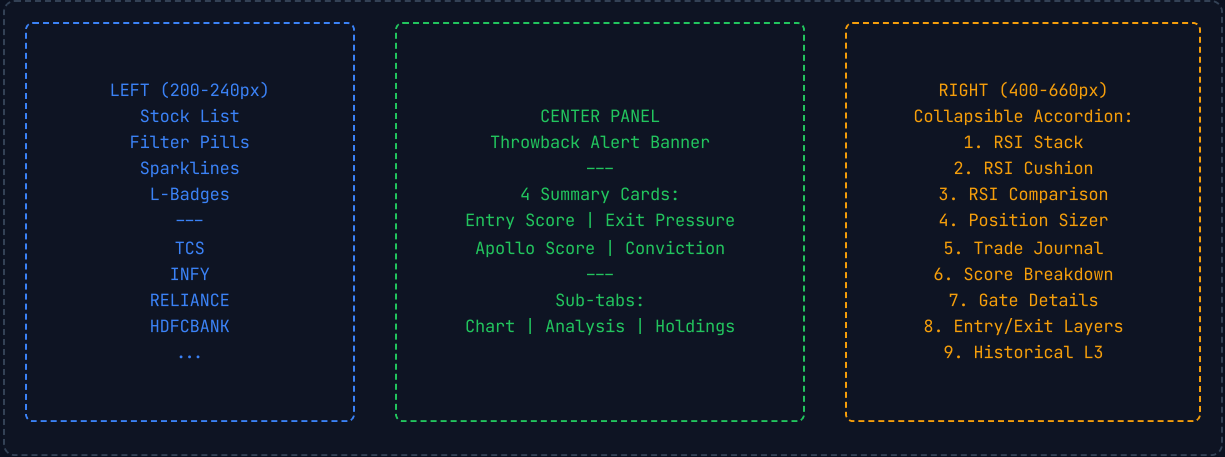
FEATURE	UI ELEMENT	LOCATION
Universe Scanning	Funnel bar + main table	Above and center of Screener
Multi-Stage Filtering	Left filter sidebar	Screener left panel
Composite Score Display	Score column with color tiers	Table column 3
Action Signal (Entry/Exit/Hold)	Badge column	Table column 4
Bucket Classification	Badge column	Table column 5
RSI Triple Display	3 color-coded numeric cells	Table columns 11-13
52-Week Position Bar	ProximityBar component	Table column 16
Gate Status Overview	Micro-dot cluster	Table column 9
Sparkline Preview	Sparkline component	Table column 20
Fundamental Quality	FQS badge	Table column 19
Row Expansion (Pool + Gates)	Expandable row panel	Below clicked row
Column Sorting	SortableTable header click	All sortable columns
Quick Filter Presets	Preset buttons in sidebar	Screener sidebar bottom

CHAPTER 6

Tab 2 -- Watchlist (PO)

The Watchlist is the deep-dive view for monitored stocks, directly adopted from the LayerSignal v21 three-column layout. It is where a trader spends the most time per stock, analyzing RSI stacks, gate details, trade journals, and position sizing. The layout divides the viewport into three panels: a narrow stock list on the left, a primary analysis panel in the center, and a collapsible detail accordion on the right.

Three-Column Layout Wireframe



Left Panel -- Stock List

The left panel is 200-240px wide and displays the user's watchlist stocks as a scrollable list. Each stock entry shows the symbol in bold, a 30-day sparkline, and an L-bucket badge (L1-L4). Filter pills at the top allow quick filtering by bucket, action status, or custom tags. The currently selected stock is highlighted with a blue left border. This panel is always visible and acts as the stock selector for the center and right panels.

Center Panel -- Primary Analysis

The center panel is the main workspace. At the top, a conditional Throwback Alert Banner appears only when the selected stock is exhibiting throwback behavior (price returning to a recent breakout level). Below the banner, four summary StatCards are arranged in a 2x2 grid: Entry Score (Apollo composite, 0-100), Exit Pressure (percentage indicating exit signal strength), Apollo Score (backtest-validated score from profile analytics), and Conviction (0.0-1.0 confidence indicator). Below the stat cards, three sub-tabs provide different views: Chart (candlestick with overlays), Analysis (text-based signal breakdown), and Holdings (current position details if any).

Right Panel -- Detail Accordion (9 Sections)

The right panel is 400-660px wide and collapsible (it can be toggled to give more space to the center panel). It contains 9 AccordionPanel sections that expand and collapse independently. These sections are the core analytical toolkit and represent the deepest level of progressive disclosure in the dashboard.

#	ACCORDION SECTION	CONTENT	SOURCE
1	RSI Stack	RSI21, RSI36, RSI56 values with historical bands and overbought/oversold zones	LayerSignal
2	RSI Cushion	Distance from RSI to overbought (70) and oversold (30) levels per period	LayerSignal
3	RSI Comparison	Side-by-side comparison of RSI values across 3 timeframes with divergence detection	LayerSignal
4	Position Sizer	Risk-based position size calculator using ATR and account risk parameters	Apollo
5	Trade Journal	Historical trade log for this stock with entry/exit dates, prices, and outcomes	Apollo
6	Score Breakdown	Individual component scores that compose the composite entry score	Apollo
7	Gate Details	All 5 gates with pass/fail, values, and threshold explanations	Apollo
8	Entry/Exit Layers	Multi-layer entry and exit price levels with probability-weighted targets	Apollo
9	Historical L3	Past instances when this stock reached L3 bucket, with outcome tracking	LayerSignal

Feature-to-UI Mapping

FEATURE	UI ELEMENT	LOCATION
Watchlist Stock Selection	Left panel stock list	Watchlist left panel
Quick Bucket Filtering	Filter pills	Left panel top
Throwback Alert	Conditional banner	Center panel top

Summary Score Cards	4x StatCard grid	Center panel below banner
Candlestick Chart	Chart sub-tab	Center panel sub-tabs
RSI Stack Visualization	Accordion section 1	Right panel
RSI Cushion Analysis	Accordion section 2	Right panel
Position Sizer	Accordion section 4	Right panel
Trade Journal	Accordion section 5	Right panel
Gate Detail Inspector	Accordion section 7	Right panel
Entry/Exit Layer Planning	Accordion section 8	Right panel
Historical L3 Tracking	Accordion section 9	Right panel

CHAPTER 7

Tab 3 -- Scanner (P1)

The Scanner tab provides a complementary view to the Screener. While the Screener shows all signal-bearing stocks with their full data profiles, the Scanner focuses on specific pattern detection across the entire market. It uses predefined pattern presets that encapsulate common trading setups, allowing traders to quickly find stocks matching specific technical patterns without manually configuring filter combinations.

Scanner Layout Wireframe



Pattern Presets

Eight predefined pattern presets cover the most common trading setups in the Apollo-LayerSignal system. Each preset encapsulates a specific combination of indicator thresholds and conditions. The L3 Ready preset finds stocks approaching the L3 bucket (strong bullish momentum). The L2 Breakout preset identifies stocks breaking out from the L2 consolidation zone. The Throwback preset finds stocks pulling back to support after a breakout. The Momentum preset catches high-ADX stocks with strong directional movement. The Vol Breakout preset detects sudden expansion in ATR. The MA Cross preset finds stocks where short and long moving averages are crossing. The Fresh L3 preset shows stocks that just entered L3 today. The Oversold preset identifies heavily over-sold conditions across all three RSI periods.

Stage Summary Bar

Above the result table, a horizontal summary bar shows the count of stocks matching each pattern preset in real time. This gives traders an instant market overview: a high count in L3 Ready suggests many stocks are approaching strong buy zones, while a high Vol Breakout count suggests the market is entering a period of increased volatility. Clicking any count in the summary bar switches to that preset and filters the table.

Feature-to-UI Mapping

FEATURE	UI ELEMENT	LOCATION
Pattern Preset Selection	Preset buttons in sidebar	Scanner left panel
Custom Filter Controls	Sliders and selectors	Scanner sidebar below presets
Pattern Match Summary	Stage summary bar	Above result table
Pattern Result Table	SortableTable	Scanner right content
Quick Watchlist Add	Row action button	Table row hover

CHAPTER 8

Tab 4 -- Analytics (PO)

The Analytics tab is the Apollo engine's primary showcase, presenting backtesting results, behavioral profiles, walk-forward validation, cross-reports, and fundamental data in five sub-tabs. This is where the quant-oriented trader spends time validating strategies, reviewing historical performance, and understanding the statistical properties of the trading system.

Sub-Tab Structure

Trade Engine

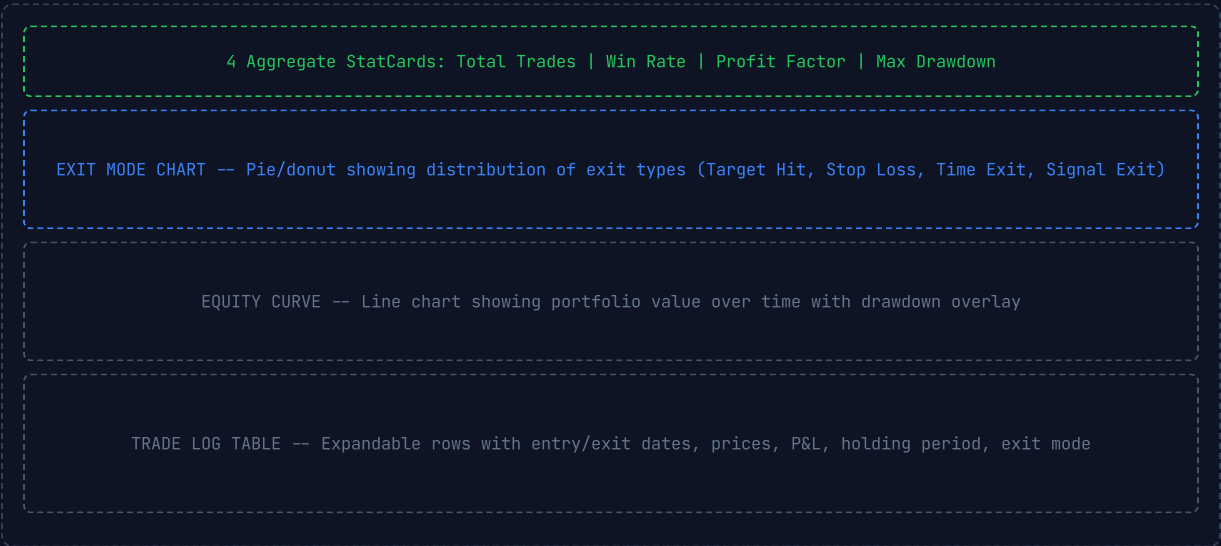
Profiles

Walk-Forward

Cross-Report

Fundamentals

Trade Engine Sub-Tab



The Trade Engine sub-tab opens with four aggregate StatCards at the top: Total Trades (integer count), Win Rate (percentage with color coding), Profit Factor (ratio with benchmark comparison), and Max Drawdown (percentage with severity indicator). Below the stats, an Exit Mode chart shows the distribution of exit types as a donut chart, helping traders understand whether the system tends to exit via targets, stop losses, time-based exits, or signal reversals. The Equity Curve shows the portfolio value over the full backtest period with a drawdown overlay shaded in red. Finally, the Trade Log is a sortable table with expandable rows showing detailed trade information.

Profiles Sub-Tab

The Profiles sub-tab displays behavioral profile reports generated by Apollo's profile_analytics.py module. Each stock in the watchlist has a profile that captures its typical post-entry behavior: average holding period, most common exit mode, MAE (Maximum Adverse Excursion) distribution, MFE (Maximum Favorable Excursion) distribution, and win rate by bucket. Profiles are displayed as card grids with each stock's profile in a dedicated card, enabling cross-stock comparison.

Walk-Forward Sub-Tab

The Walk-Forward sub-tab presents the results of Apollo's walk-forward validation engine. This tab shows how the trading system performs on out-of-sample data across multiple rolling windows. Key metrics include rolling win rates, stability scores (variance of performance across windows), and parameter sensitivity analysis. The tab uses a combination of line charts (for rolling metrics over time) and summary tables (for aggregate statistics). This is critical for verifying that the system is not overfit to historical data.

Feature-to-UI Mapping

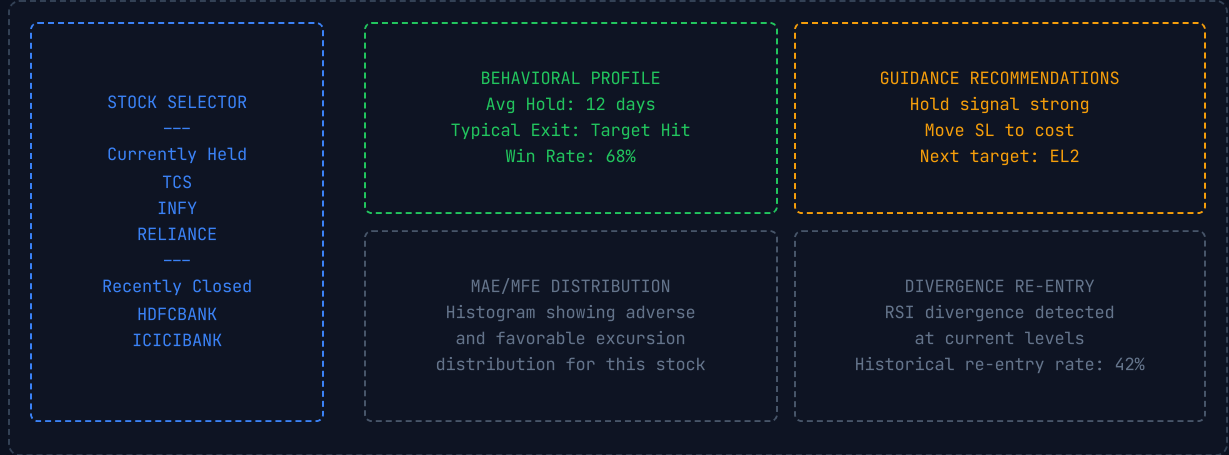
FEATURE	UI ELEMENT	SUB-TAB
Aggregate Trade Statistics	4x StatCard grid	Trade Engine
Exit Mode Distribution	Donut chart	Trade Engine
Equity Curve	Line chart with drawdown	Trade Engine
Trade Log with Details	SortableTable with expand	Trade Engine
Behavioral Profile Cards	Card grid per stock	Profiles
MAE/MFE Distributions	Histogram charts	Profiles
Win Rate by Bucket	Bar chart / table	Profiles
Profile Comparison	Side-by-side cards	Profiles
Walk-Forward Rolling Metrics	Multi-line chart	Walk-Forward
Stability Score Display	StatCard with gauge	Walk-Forward
Parameter Sensitivity	Heatmap table	Walk-Forward
Cross-Report Summary	Combined stats table	Cross-Report
Regime-Adjusted Performance	Grouped bar chart	Cross-Report
Fundamental Data Table	SortableTable	Fundamentals
Score vs. Fundamentals Scatter	Scatter plot	Fundamentals

CHAPTER 9

Tab 5 -- Guidance (P1)

The Guidance tab provides post-trade behavioral analysis and actionable recommendations. It is designed for traders who want to understand not just what to trade, but how to manage positions based on historical behavior patterns. This tab draws heavily from Apollo's guidance engine, which analyzes past trades to generate probabilistic recommendations for current positions.

Guidance Layout Wireframe



Behavioral Profile Section

The Behavioral Profile card shows the selected stock's historical trading behavior based on Apollo's `profile_analytics.py` output. It displays the average holding period, the most common exit mode, the win rate for this specific stock, and how it compares to the system-wide average. This section answers the question: "When I trade this stock, what typically happens?" It helps traders set realistic expectations and adjust position sizes based on historical behavior.

Guidance Recommendations

The Recommendations card provides specific, actionable guidance for the current position (or the most recent position if none is active). Recommendations are generated by Apollo's guidance engine and include suggestions like "hold signal remains strong," "consider moving stop-loss to cost," or "next target level is EL2 based on historical layer progression." Each recommendation is tagged with a confidence level derived from the historical frequency of that recommendation leading to a positive outcome.

MAE/MFE Distribution

The MAE/MFE Distribution section shows histograms of Maximum Adverse Excursion and Maximum Favorable Excursion for the selected stock. MAE tells the trader how much the stock typically moves against them before reversing (informing stop-loss placement), while MFE tells how much the stock typically moves in their favor before reversing (informing target placement). These distributions are generated from the `mae_mfe.py` module and provide statistically grounded position management guidance.

Divergence Re-Entry Analytics

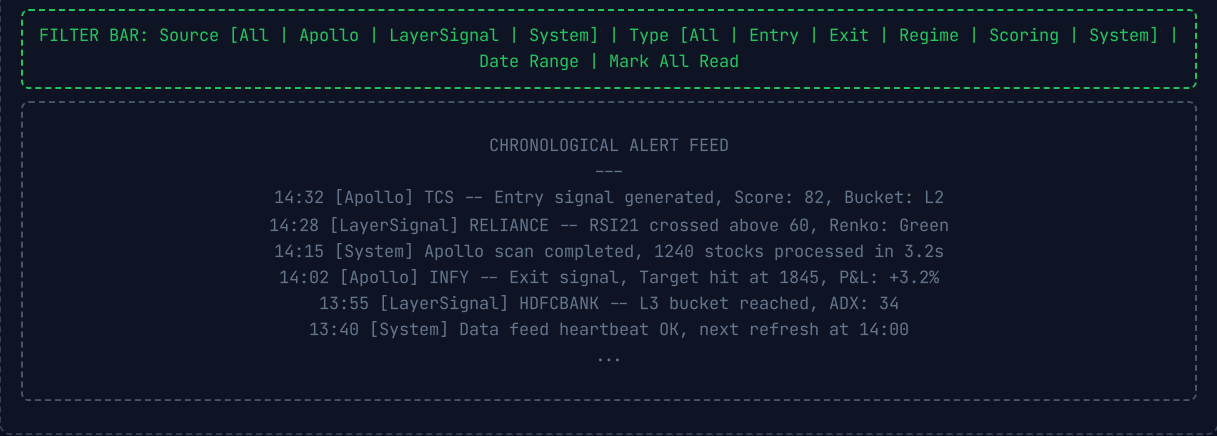
This section detects RSI divergence patterns at the current price level and compares them against historical instances where similar divergences occurred. It shows the historical re-entry success rate (percentage of times a divergence led to a profitable re-entry) and the typical holding period after re-entry. This is particularly useful for traders who exited a position and are evaluating whether to re-enter at the current level.

CHAPTER 10

Tab 6 -- Alerts (PO)

The Alerts tab is the chronological notification center for the entire system. It aggregates alerts from both the Apollo and LayerSignal engines, as well as system-level notifications, into a single unified feed. Unlike traditional dashboards that scatter alerts across multiple views, this design centralizes all notifications in one place while allowing filtering by source, type, and time range.

Alerts Layout Wireframe



Alert Sources

Three alert sources populate the feed. Apollo alerts include entry signal generation, exit signal triggers (target hit, stop loss, time exit), scoring changes, and gate status transitions. LayerSignal alerts include bucket changes (especially L2 to L3 transitions), RSI threshold crossings, Renko brick changes, and throwback pattern detections. System alerts include engine scan completion notifications, data feed health checks, and error conditions. Each alert in the feed displays its source as a colored prefix badge (purple for Apollo, green for LayerSignal, teal for System).

Alert Types and Interaction

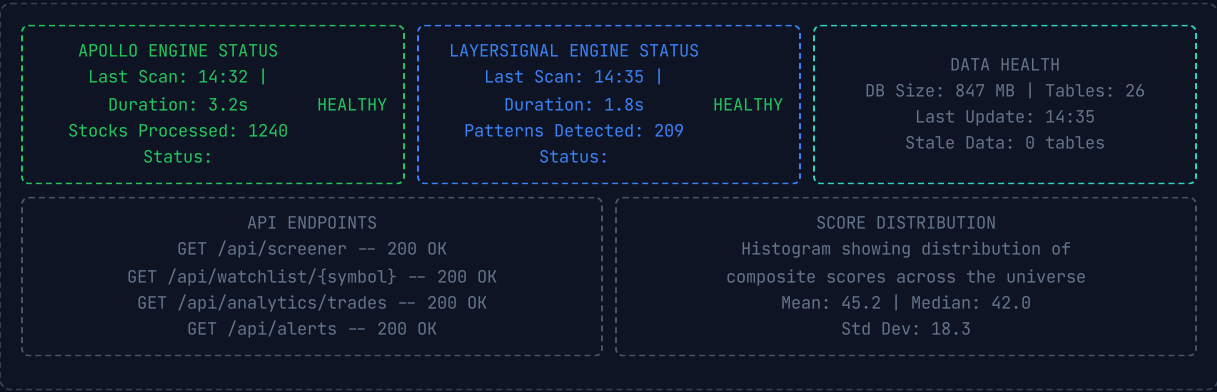
Alerts are categorized into five types: Entry (new trading opportunity), Exit (position closure signal), Regime (market regime change), Scoring (significant score movement), and System (operational status). The filter bar at the top allows combining source and type filters for precise alert isolation. Each alert card is clickable, navigating to the relevant stock in the Watchlist tab or the relevant section in the System tab. Unread alerts show a blue left border indicator. The feed auto-scrolls to the latest alert but preserves the user's scroll position when they are reading historical alerts.

CHAPTER 11

Tab 7 -- System (P2)

The System tab provides operational visibility into the engines, data pipelines, and overall health of the dashboard. It is the lowest-priority tab (P2) because it is not part of the daily trading workflow, but it is essential for debugging, maintenance, and ensuring data quality. The layout uses a card grid pattern to present status information clearly.

System Layout Wireframe



Engine Status Cards

Three primary status cards show the real-time health of each engine and the data layer. The Apollo Engine card displays the timestamp and duration of the last scan cycle, the number of stocks processed, and a health indicator (green/amber/red). The LayerSignal Engine card shows equivalent information for the live scanning engine. The Data Health card shows the SQLite database size, the number of tables (26), the timestamp of the last data update, and a count of any tables with stale data (data older than the expected refresh interval). Any card showing amber or red status can be clicked to reveal detailed diagnostic information.

Detail Cards

Below the status cards, two detail cards provide additional operational context. The API Endpoints card lists all REST API endpoints served by the FastAPI backend, their current response status, and average response time. This helps identify any API degradation. The Score Distribution card shows a histogram of composite scores across the entire universe, along with statistical summaries (mean, median, standard deviation). This helps identify whether the scoring system is producing well-distributed outputs or if there is skew that might indicate a calibration issue.

CHAPTER 12

Cross-Cutting Design Patterns

This chapter defines the interaction patterns, state displays, and responsive behaviors that apply uniformly across all seven tabs. Consistency in these patterns is critical for a dashboard that will be used for hours at a time, as even minor inconsistencies in interaction behavior create cognitive friction that compounds over a trading session.

Interaction Patterns

INTERACTION	TRIGGER	RESPONSE	TIMING
Hover	Mouse over data cell	Tooltip with full value, definition, and source engine	150ms delay, 200ms fade-in
Click	Single click on row	Expand row in-place (Screener/Scanner) or select stock (Watchlist)	Instant
Double-click	Double click on symbol	Navigate to Watchlist tab with that stock selected	Instant
Expand	Click accordion header	Toggle accordion section open/closed with animation	250ms slide
Collapse	Click accordion header or panel toggle	Collapse right panel in Watchlist to give more center space	300ms slide
Sort	Click table column header	Sort table by that column, toggle asc/desc	Instant with 150ms highlight
Filter	Toggle filter pill or adjust slider	Filter table/fed in real-time without page reload	100ms debounce
Search	Type in global search	Show autocomplete dropdown with matching symbols and features	200ms debounce

Loading, Error, and Empty States

Loading states use skeleton screens that mirror the final layout structure. Rather than showing a spinner, the table rows, card grids, and chart areas display pulsing placeholder shapes that match the dimensions of the expected content. This reduces perceived load time and prevents layout shift when data arrives. Error states appear inline (not as modal dialogs) with a red-tinted bar at the top of the affected section, showing the error message and a retry button. System-level errors also push an alert to the Alerts tab. Empty states show a centered illustration with a brief message explaining why no data is displayed (for example, "No stocks match the current filter" or "No back-test results available for the selected date range") along with a suggested action button.

Toast notifications appear in the bottom-right corner for transient events: successful data refresh, alert count changes, and background task completions. Toasts auto-dismiss after 4 seconds with a manual dismiss option.

They are non-blocking and stack vertically if multiple toasts are active simultaneously. The toast uses the semantic color of the event: green for success, amber for warning, red for error, and blue for informational.

Responsive Behavior

The dashboard is optimized for two primary breakpoints. At 1920px (full HD), all three Watchlist panels are visible with comfortable spacing, the Screener shows all 20+ columns without horizontal scrolling, and charts render at full detail. At 1366px (common laptop resolution), the Watchlist right panel auto-collapses and can be toggled, the Screener table enables horizontal scrolling for columns beyond the viewport, and charts reduce their detail level. The dashboard does not support mobile or tablet viewports as trading dashboards require precise data reading that is impractical on small screens.

CHAPTER 13

Architecture-to-UI Requirement Mapping

This chapter provides a comprehensive mapping from the backend engine outputs (Python modules and JavaScript code) to their specific UI locations in the dashboard. This mapping serves as the implementation bridge between the engineering architecture and the front-end design. Each row identifies the source module, the specific data it produces, and the exact UI element that consumes it.

scoring.py Outputs

OUTPUT	UI LOCATION	TAB
Composite score (0-100)	Score column in Screener table	Screener
Individual component scores	Score Breakdown accordion	Watchlist
Action signal (Entry/Exit/Hold)	Action badge column	Screener, Watchlist
Conviction score	Conviction MetricBadge	Screener, Watchlist

trade_engine.py Outputs

OUTPUT	UI LOCATION	TAB
Trade log (entry/exit dates, prices, P&L)	Trade Log table with expandable rows	Analytics
Aggregate statistics (win rate, profit factor)	4x StatCard grid	Analytics
Exit mode distribution	Exit Mode donut chart	Analytics
Equity curve data	Equity Curve line chart	Analytics

profile_analytics.py Outputs

OUTPUT	UI LOCATION	TAB
Behavioral profile (avg hold, typical exit, win rate)	Profile card in card grid	Analytics (Profiles)
MAE/MFE distributions	Histogram charts	Analytics (Profiles), Guidance
Win rate by bucket	Bar chart	Analytics (Profiles)

walk_forward_validator.py Outputs

OUTPUT	UI LOCATION	TAB
Rolling window metrics	Multi-line chart	Analytics (Walk-Forward)
Stability score	StatCard with MiniGauge	Analytics (Walk-Forward)
Parameter sensitivity matrix	Heatmap table	Analytics (Walk-Forward)

cross_report_pipeline.py Outputs

OUTPUT	UI LOCATION	TAB
Combined Apollo + LayerSignal report	Cross-Report stats table	Analytics (Cross-Report)
Market regime label	Market Regime Banner	Global (all tabs)
Regime-adjusted performance	Grouped bar chart	Analytics (Cross-Report)
Unified signal summary	Summary cards in Watchlist	Watchlist

indicators.py Outputs

OUTPUT	UI LOCATION	TAB
RSI values (21, 36, 56 period)	RSI color-coded columns	Screener, Watchlist, Scanner
ADX value	ADX data column	Screener, Scanner
ATR and ATR% value	ATR% data column	Screener, Watchlist, Scanner
52-week position data	ProximityBar component	Screener, Watchlist
Renko brick data	Renko badge	Screener, Watchlist, Scanner

LayerSignal JS Engine Outputs

OUTPUT	UI LOCATION	TAB
Bucket classification (L1-L4)	Bucket badge column	Screener, Watchlist
Pattern detection results	Scanner result table	Scanner
Live RSI stack data	RSI Stack accordion	Watchlist
Throwback pattern alerts	Throwback Alert Banner	Watchlist

Both Engines -- Shared Outputs

OUTPUT	UI LOCATION	TAB
Alert events (entry, exit, regime, scoring)	Chronological alert feed	Alerts
Engine health status	Engine status cards	System
Data timestamps	Timestamps throughout UI	All tabs

CHAPTER 14

Design Decisions

This chapter documents the eight key design decisions made during the creation of this specification. Each decision is presented as a card with its title, the rationale behind the choice, and whether it was adopted directly from an existing source (Apollo architecture document or LayerSignal v21) or adapted with modifications. These decisions represent the points where this specification intentionally diverges from or extends its source materials.

1. Task-Based Tabs (Not Engine-Based)

Status: Departure from Architecture Document

The architecture document proposed an engine-mode toggle that would switch the entire dashboard between Apollo and LayerSignal views. This specification rejects that approach in favor of task-based tabs (Screener, Watchlist, Scanner, etc.). Rationale: Traders think in terms of tasks, not engines. A trader wants to "check my watchlist" or "scan for new opportunities," not "use Apollo." The task-based structure also enables seamless mixing of both engines' outputs in a single view (e.g., the Screener table showing Apollo scores alongside LayerSignal RSI values).

2. Three-Column Watchlist Layout

Status: **Adopted** from LayerSignal v21

The three-column layout (stock list, center panel, detail accordion) is directly adopted from the existing LayerSignal v21 interface. This decision was made because the current LayerSignal layout already represents months of iterative refinement based on daily usage feedback. The right panel is made collapsible (not fixed) to accommodate smaller screens, which is an adaptation from the original fixed-width design.

3. Combined Screener Table

Status: **Adapted** from both sources

The architecture document described separate Apollo and LayerSignal screener views. This specification combines them into a single table with 20+ columns. Rationale: Splitting the screener by engine would force traders to check two separate views and mentally merge the results. A combined table with sortable columns lets traders see the full picture in one place. The column ordering follows the LayerSignal convention (RSI values in green/blue/amber) while adding Apollo columns (score, gates, conviction) to the right.

4. RSI Color Lock

Status: **Adopted** from LayerSignal v21

The RSI color assignments (RSI21=green, RSI36=blue, RSI56=amber) are locked and must not be changed. These colors are burned into trader muscle memory from months of LayerSignal usage. Any change would cause immediate confusion and slow down signal recognition. This is a non-negotiable constraint that overrides any aesthetic considerations.

5. Dark Blue Theme

Status: **Adapted** from LayerSignal dark mode

LayerSignal used a generic dark gray theme. This specification adapts it to a dark blue theme (using the #0B1120 to #475569 range from Tailwind's slate palette). Rationale: Blue undertones reduce eye strain more effectively than neutral gray during extended screen sessions, and they provide a distinctive visual identity that differentiates this dashboard from the many gray-dark trading interfaces in the market. The blue theme also creates better contrast with the green/red signal colors.

6. Badge-First Visual Grammar

Status: **Adopted + Expanded** from LayerSignal v21

LayerSignal introduced the concept of color-coded badges for bucket classification (L1-L4). This specification expands the badge system to cover all categorical data: action signals (Entry/Exit/Hold), FQS grades (A/B/C/D), entry layers (EL1-EL4), Renko states, and alert types. The badge grammar is consistent: green for bullish/positive, red for bearish/negative, amber for caution/neutral, blue for informational, and purple for Apollo-specific data.

7. Flags Inform, Never Gate

Status: **Adopted** from Architecture Document

The architecture document established the principle that binary flags (such as IPO restriction, regime mismatch, or data staleness) inform the trader but never prevent them from seeing a signal. This specification enforces this principle by displaying flags as informational FlagChip components (small on/off indicators) rather than using them as filter gates. A stock with an active IPO flag will still appear in the Screener table with the flag visible, allowing the trader to make an informed decision rather than having the system silently suppress it.

8. Footer Ticker Tape

Status: **Adapted** from LayerSignal v21

The scrolling ticker tape at the bottom of the viewport is adopted from LayerSignal v21. This specification expands it from the original 8 indices to 14 indices for broader market coverage. The ticker provides constant market context without requiring the trader to navigate away from their current tab. It uses the NSE data feed engine for real-time index values and auto-scrolls horizontally when the content exceeds the viewport width.

CHAPTER 15

Implementation Priority

The 101 features are organized into four implementation phases based on priority and dependency. Phase A is the minimum viable dashboard and must be completed first. It contains all P0 features across the four P0 tabs (Screener, Watchlist, Analytics, and Alerts) plus the global shell components. Phases B through D add progressively lower-priority features that enhance the dashboard but are not required for initial deployment.

Phase A -- Minimum Viable Dashboard (46 features)

Phase A implements the four P0 tabs (Screener, Watchlist, Analytics, Alerts) with their core features, plus the global shell (App Bar, Tab Bar, Regime Banner, Ticker Footer). This phase delivers a fully functional trading dashboard that covers the primary daily workflow: scan the universe for signals, deep-dive into watchlist stocks, review backtest analytics, and monitor alerts. The combined Screener table, three-column Watchlist, Trade Engine sub-tab, and unified alert feed are the centerpiece deliverables of this phase.

P0 -- 46 features

Screener: 13 features

Watchlist: 12 features

Analytics: 15 features

Alerts: 6 features

Phase B -- Enhanced Workflow (33 features)

Phase B adds the two P1 tabs (Scanner and Guidance) with their full feature sets. The Scanner provides pattern-based market scanning with 8 presets, complementing the Screener's score-based approach. The Guidance tab adds post-trade behavioral analysis and recommendations, transforming the dashboard from a signal display tool into a complete trading decision support system. This phase also includes enhanced interaction patterns like advanced filtering, saved views, and keyboard shortcuts.

P1 -- 33 features

Scanner: 5 features

Guidance: 4 features

Cross-cutting: 24 features

Phase C -- Operational Tooling (19 features)

Phase C implements the P2 System tab with engine health monitoring, API endpoint status, and score distribution analytics. It also includes data export capabilities (CSV and PDF report generation), advanced configuration options, and performance optimization features. This phase is essential for long-term maintainability and operational visibility but is not required for the initial trader-facing deployment.

P2 -- 19 features

System: 8 features

Export/Config: 11 features

Phase D -- Future Enhancements (3 features)

Phase D contains P3 features that are deferred to a future roadmap. These include multi-monitor support for traders running multiple screens, advanced collaboration features (shared watchlists and annotations), and a mobile-friendly read-only view for checking alerts on the go. These features are documented here for completeness but are explicitly out of scope for the initial implementation.

P3 -- 3 features

Phase Summary

Phase	Priority	Features	Key Deliverables
A	P0	46	Screener, Watchlist, Analytics, Alerts, Global Shell
B	P1	33	Scanner, Guidance, Advanced Interactions
C	P2	19	System Tab, Export, Configuration
D	P3	3	Multi-monitor, Collaboration, Mobile Read-only
Total		101	Complete Unified Dashboard

101 features.

7 tabs.

26 SQLite tables.

2 engines.

1 unified interface.